

👉 XAI Lecture 14

Network Dissection & TCAV

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

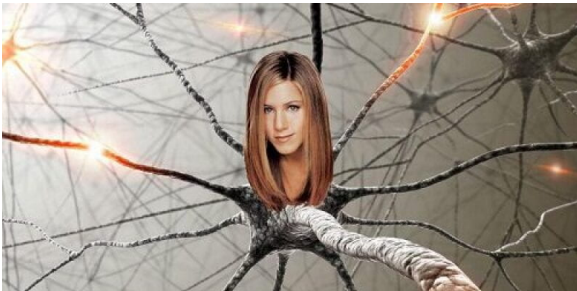
Network Dissection: Quantifying Interpretability of Deep Visual Representations

Authors: David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, Antonio Torralba (CSAIL, MIT)

Presented by: Anat Kleiman, Gustaf Ahdritz, Xin Tang, Luke Bailey

[@bau2017network]

🧠 How Is Semantic Visual Concept Represented in the Brain? ---



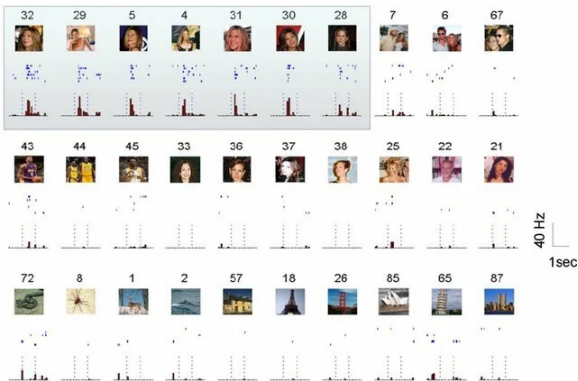
Neuroscientists have found that individual neurons can respond selectively to specific concepts — a property called **selectivity**.

The debate

Does the brain use **local** representations (one neuron = one concept) or **distributed** ones (concepts spread across many neurons)?

Question: Do deep neural networks learn similar structure?

🍷 A Neuron That Only Fires for Jennifer Aniston ---



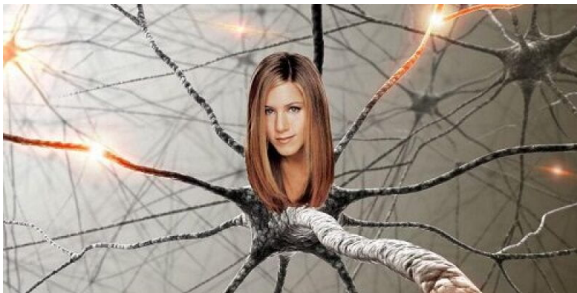
Quiroga et al. (2005) recorded neurons in the human medial temporal lobe and found neurons that fire exclusively for specific people or landmarks.

- One neuron fired for Jennifer Aniston — photos, drawings, even her name written in text
- Another neuron fired only for the Eiffel Tower

Why this matters

This is a **disentangled** representation: one neuron = one concept. The paper asks whether CNNs learn something analogous.

🍷 Disentangled Representation in Visual Cortex ---



The visual cortex processes information in a **hierarchy** — the **ventral stream** (“what pathway”):

- **V1**: edges, orientations (small receptive fields)
- **V2/V4**: textures, shapes, color patterns
- **IT** (inferotemporal): objects, faces — where the “Jennifer Aniston neuron” lives

Along this hierarchy: **selectivity** increases and **invariance** increases.

Bridge to CNNs

This layered specialization is the biological inspiration for deep CNNs. Network Dissection tests whether CNNs develop analogous detectors at each layer.

🍷 Disentangled Representation in Visual Cortex ---

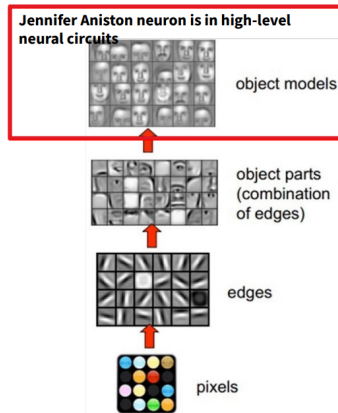
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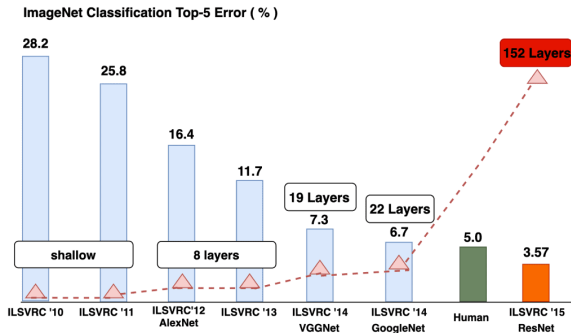
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Bridge to CNNs

This layered specialization is the biological inspiration for deep CNNs. Network Dissection tests whether CNNs develop analogous detectors at each layer.



🍷 Deep CNN for Computer Vision ---



CNNs revolutionized computer vision, reaching superhuman performance on ImageNet — but they remain largely opaque.

- Units in early layers respond to edges and colors
- Units in later layers seem to respond to objects and scenes (Zhou et al., 2015)
- But this was observed qualitatively — no rigorous measurement existed

The gap

We know CNNs *work* extremely well. We don't know *how*, or what each unit has learned. Network Dissection provides the first quantitative answer.

Proposed Questions ---

The paper frames the problem around three concrete research questions. Each has a direct answer by the end of the paper.

- ❶ **What is a disentangled representation, and how can its factors be quantified and detected (in deep CNN)?**
- ❷ **Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?**
- ❸ **What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?**

Proposed Questions and Contributions ---

Q1: How to quantify disentangled representations?

- Proposed a metric, intersection over union score (IoU), to quantify the interpretability of each unit
- The alignment level between unit activated area and human-interpretable concepts

Q2: Are interpretable units real or a chimera?

- A semantic concept can be detected by many units
- A unit can detect many semantic concepts
- But the natural basis is **special**: rotating it destroys interpretability

Q3: What training conditions affect entanglement?

- Factors tested: layer depth, training iterations, the angle of the images, input datasets, dropout, batch normalization, supervised vs. self-supervised, layer width

🍷 Related Works ---

Prior work tried to understand CNN internals through **visualization**, but all approaches share a fundamental limitation: they are qualitative and cannot be used to rigorously compare models.

- **Generative Visualizations**: Synthesize images that maximally activate a unit (Mahendran et al., 2015; Nguyen et al., 2016; Simonyan et al., 2014)
- **Salience-based Visualizations**: Highlight which pixels most contribute to a unit's activation (Zeiler et al., 2014)
- **Global Analysis**: Project the full representation space into 2D for inspection (t-SNE: van der Maaten et al., 2008; Yosinski et al., 2015)

Common limitation

All these methods produce images that a **human** must then interpret — subjective and impossible to compare across networks. Network Dissection replaces the human with a **metric**.

🍌 Broden: Broadly and Densely Labeled Dataset ---

To measure interpretability we need a **ground truth** of human concepts. Broden unifies five existing segmentation datasets into a single resource.

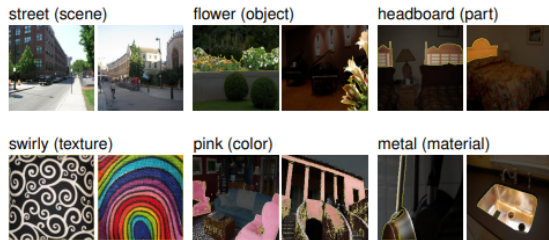


Figure 2. Samples from the **Broden** Dataset. The ground truth for each concept is a pixel-wise dense annotation.

Category	# Classes
Scene	468
Object	584
Part	234
Material	32
Texture	47
Color	11

Multiple labels can apply to the **same pixel** (e.g., a black cat leg → “cat”, “leg”, “black”)

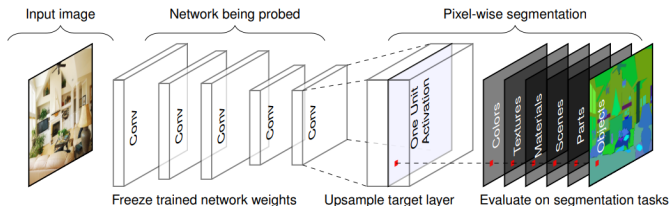
The range from colors to scenes allows testing nterpretability at **every level of abstraction**.

🍷 Scoring Unit Interpretability ---

Pipeline:

- Feed images from Broden through the frozen CNN
- Collect activation map $A_k(x)$ for every unit k
- Upsample to input resolution $\rightarrow S_k(x)$
- Threshold to obtain a binary mask $M_k(x)$
- Compare against concept labels L_c

Each convolutional unit k acts like a **filter** that fires on certain spatial regions of an image. The core idea: treat each unit as a candidate **segmentation model** for some concept. For each unit, the method evaluates its activation mask against all **1,197 segmentation tasks** in Broden (one per concept). The unit is assigned the concept with the highest score.



🍷 Scoring Unit Interpretability: Threshold ---



threshold

Threshold T_k : value such that
 $P(a_{ij}^k > T_k) = 0.005$

This means: only the **top 0.5%** of activations for unit k across the entire dataset are considered “active.”

Why a quantile?

Each unit has a different activation range. A fixed threshold would favor high-magnitude units. The quantile normalizes this.

🍷 Scoring Unit Interpretability: IoU ---



$$IoU_{k,c} = \frac{\sum_x |M_k(x) \cap L_c(x)|}{\sum_x |M_k(x) \cup L_c(x)|}$$

In plain English

“Of all the pixels where **either** the unit fires **or** the concept is present, what fraction has **both**?”

Detector definition

Unit k is a detector for concept c if $IoU_{k,c} > 0.04$

🍲 Quantifying Alignment: Summary ---

The interpretability of unit k for concept c is the **IoU** across the dataset:

- **Detector**: Unit k is a detector for concept c if $IoU_{k,c} > 0.04$
- **Top Label**: If a unit matches multiple concepts, the one with the highest IoU is assigned
- **Layer Score**: The interpretability of a layer is the number of **unique** semantic concepts aligned with its units

Why "unique"?

If 50 units all detect "dog", the layer score counts "dog" only once. This measures **diversity** of concepts, not just quantity of detectors.

Experiments: Tested CNN Models ---

The paper evaluates interpretability across diverse architectures and training regimes to identify which conditions favor disentangled representations.

Training	Network	Dataset / Task
None	AlexNet	Random weights
Supervised	AlexNet, GoogLeNet, VGG-16, ResNet-152	ImageNet, Places205, Places365, Hybrid
Self-supervised	AlexNet	10 proxy tasks (context, puzzle, colorization, etc.)

Places: scene-centric dataset with categories such as kitchen, living room, coast.

Surprising finding (preview)

Models trained on **Places** (scenes) develop more *object* detectors than those trained on **ImageNet** (objects). Why? Because recognizing a scene *requires* identifying the objects in it.

🍷 Experiment 1: Human Evaluation of Interpretations ---

- Does Network Dissection agree with **human judgment**?
 - ▶ Raters shown 15 images with highlighted patches for each unit in AlexNet (Places205)
 - ▶ Asked: “Does this phrase describe most of the patches?” (yes/no)
 - ▶ **Network Dissection accuracy**: portion of method’s labels rated as descriptive
 - ▶ **Human consistency**: portion of ground-truth labels found descriptive by a second group of raters

	conv1	conv2	conv3	conv4	conv5
Interpretable units	57/96	126/256	247/384	258/384	194/256
Human consistency	82%	76%	83%	82%	91%
Network Dissection	37%	56%	54%	59%	71%

Two takeaways

1. Network Dissection agreement **increases** in higher layers (37% → 71%)
2. Human consistency remains high throughout (76–91%), setting an upper bound

🍷 Experiment 2: Axis-Aligned Interpretability ---

Is interpretability a property of the **natural basis** learned by the network, or does it appear in **every direction**?

Hypothesis 1: Concepts appear in every direction

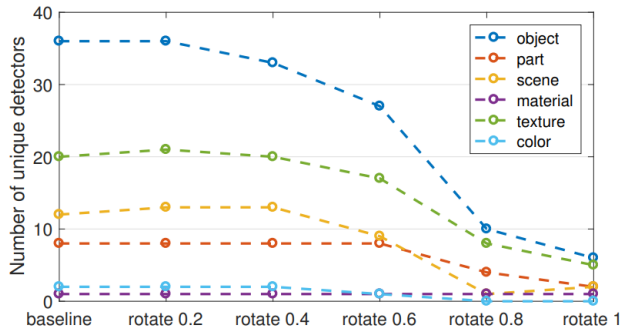
- Single units are not more interpretable than random combinations
- Interpretability would be an artifact, not a real property

Hypothesis 2: The natural basis is special

- The model converges to a semantically rich, axis-aligned basis
- Interpretability is a learned property of the representation

Test: Apply a random orthogonal rotation R to the representation space. If H2 is correct, rotating the basis should **destroy** interpretability.

🍷 Experiment 2: Rotation Results ---

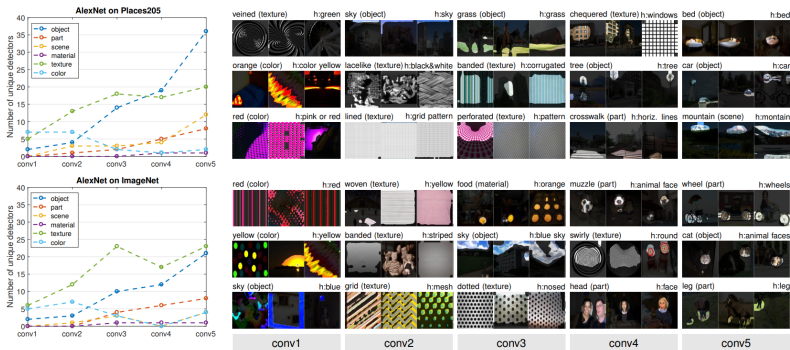


Number of unique detectors **decreases up to 80%** as the basis is rotated away from the natural one.

Conclusion

Interpretability is **not** an inevitable byproduct of discriminative power. It is a **special alignment** learned by the network — **Hypothesis 2 confirmed.**

🍷 Experiment 3: Concepts by Layer ---



The type of concept detected changes with **layer depth**: **Early layers**: colors, textures → **Middle layers**: materials, parts → **Late layers**: objects, scenes.

This mirrors the visual cortex hierarchy: V1 (edges) → V4 (textures) → IT (objects). The analogy holds quantitatively.



Experiment 4: Network Architectures ---

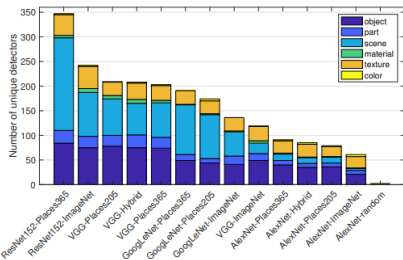
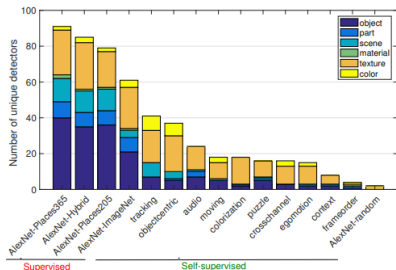


Figure 7. Interpretability across different architectures and training.



Comparing the number of unique detectors across architectures trained on the **same dataset**:

- Deeper and more complex architectures tend to develop **more unique concept detectors**
- **ResNet-152** and **VGG-16** consistently outperform AlexNet and GoogLeNet

Key nuance

More detectors does not always mean better accuracy — the relationship is architecture-dependent.

Experiment 5: Training Conditions ---

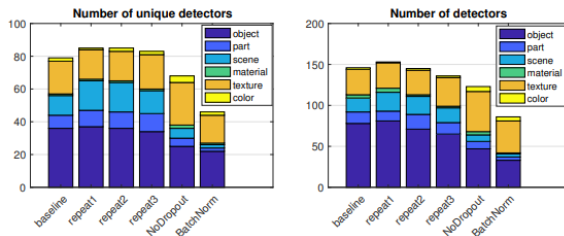
What training choices affect interpretability?

- **Weight initializations: Minimal effect** — models converge to similar levels of interpretability regardless of random seed
- **Dropout: Some effect** — removing dropout leads to more “texture” detectors and fewer “object” detectors
- **Batch Normalization: Significant effect** — interpretability decreased substantially

Surprising result

Batch Normalization improves accuracy but **hurts** interpretability. There may be a trade-off between optimization convenience and semantic disentanglement.

🍷 Experiment 5: Training Conditions Results ---



The figure confirms:

- Different random seeds → nearly identical interpretability
- Dropout removal → shift from objects to textures
- Batch Norm → large drop in unique detectors across all categories

🍷 Experiment 6: Discrimination vs. Interpretability ---

Does being more “interpretable” make a model **better at new tasks**?

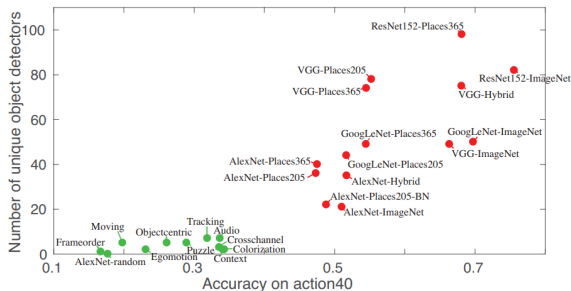
Benchmark:

- Across several CNNs, extract activations from high layers
- Train a **linear SVM** on a new task: **action recognition** (Action40 dataset)
- Compare classification accuracy vs. number of unique object detectors

Why a linear SVM?

A linear classifier can only succeed if the features are already well-organized. If interpretable representations also produce linearly separable features, that's strong evidence they generalize.

🍷 Experiment 6: Discrimination Results ---



Result: Positive correlation between object detectors and classification accuracy.

Implication

Encouraging **concept detection** can improve **discrimination**. Interpretability and performance are **not** at odds — they can reinforce each other.

Experiment 7: Layer Width ---

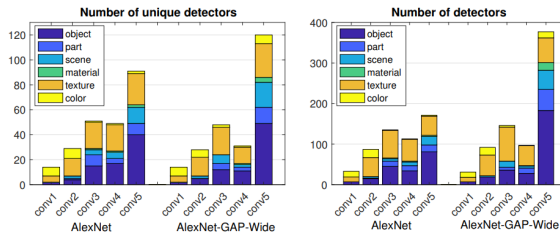
What happens if we make a layer **wider** (more units)?

- **Method:** Tripled the number of units in AlexNet's conv5 (from 256 to 768)
- **Result:** Accuracy remains similar, but the number of **unique concept detectors** increases significantly
- **Limit:** Beyond ~1024 units, diminishing returns in unique concepts

Intuition

Wider layers provide more "slots" for the network to separate explanatory factors — like having a bigger bookshelf with room for more categories.

🍷 Experiment 7: Width Results ---



Increased width:

- More unique detectors both at the widened layer **and** in the network generally
- Accuracy remains stable
- Effect saturates beyond a threshold

If interpretability is a goal, **wider layers** are a cheap way to get more disentangled representations without sacrificing accuracy.

Discussion Questions (Paper 1) ---

- ➊ **Distribution Understanding:** Concept detectors from a particular dataset betray something about the underlying distribution. How do you think this can be applied in the real world (e.g., bias detection)?
- ➋ **Single unit to circuit:** This method interprets single units; could it be extended to larger circuits and would this be useful?
- ➌ **Beyond vision:** This method is deeply tied to the vision domain. Could it be extended to other domains such as natural language?

Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)

Authors: Kim et al. (2018)

Presented by: Lucia Gordon, Matthew Nazari, Catherine Yeh

[@kim2018interpretability]

🍷 From Network Dissection to TCAV - ver si lo dejo ---

Network Dissection (Bau et al., 2017)

- Interprets **individual units**
- Concepts come from a **fixed dataset** (Broden)
- Measures: does unit k detect concept c ?
- Spatial alignment (IoU)
- Limited to CNNs with spatial activations

TCAV (Kim et al., 2018)

- Interprets **entire classes**
- Concepts defined by the **user** (any set of examples)
- Measures: is concept c important for class k ?
- Directional sensitivity in activation space
- Works on any differentiable model

The shift

Network Dissection asks: "what does this unit detect?"

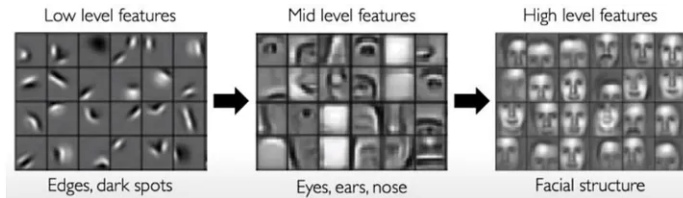
TCAV asks: "how important is this concept for this prediction?"

🍷 Motivation + Problem Statement ---

- Interpreting deep learning models is crucial to understanding their behavior and ensuring accurate predictions
- But remains a big challenge due to size, complexity, and opacity of ML models
- Many systems operate on **low-level features** (e.g., pixel values) rather than **high-level concepts** (e.g., face) that are human-interpretable

The mismatch

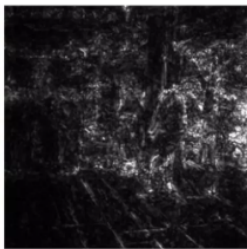
Models explain in **pixels**. Humans understand in **concepts**. TCAV bridges this gap.



🍷 Motivation + Problem Statement ---

Problem:

- We can't express these concepts as pixels
- And they weren't our input features
- Existing methods (saliency maps, LIME) explain in terms of input features — not the concepts humans care about



What we want to ask

"Did the model use the concept of *person*? Of *money*? Of *machine*?" — not "which pixels mattered?"

🍷 Summary of Contributions ---

- Introduce **Concept Activation Vectors (CAVs)**: way to interpret a neural network's internal state in terms of human-friendly concepts
- Key idea: use the high-dimensional internal state of a neural net as an **aid**, not an obstacle
- Main contribution: **Testing with CAV (TCAV)**, that quantifies model sensitivity to a high-level concept learned by a CAV for a particular class

Example

"How sensitive is the class *zebra* to the concept *striped*?" → TCAV returns a single number.



High level concept: "striped"



Particular class: "zebra"



🍷 Goals of TCAV ---

- **Accessibility:** requires little to no ML expertise
- **Customization:** adaptable to any concept, even outside of training
- **Plug-in readiness:** works without retraining/modifying ML models
- **Global quantification:** can interpret entire classes with a single quantitative measure



Assessed with experiments + human evaluation

Compare with Network Dissection

ND is not accessible (requires understanding IoU), not customizable (fixed Broden concepts), and not global (per-unit, not per-class). TCAV addresses all three.

🍷 Related Work: Limitations of Current Methods ---

Perturbation-based (LIME, SHAP):

- Explain individual predictions (**local**), not entire classes
- Operate at the input feature level, not at the concept level

Saliency methods (Gradient, CAM, etc.):

- **Local**: one image at a time
- **Lack customization**: cannot test user-defined concepts
- **Vulnerable** to adversarial attacks (Ghorbani et al., 2017)
- **Insensitive** to model randomization (Adebayo et al., 2018)

Core limitation shared by all

These methods explain in terms of **input features**. Saliency maps tell you *where* the model looks, not *what* it sees. TCAV explains in terms of **human-defined concepts**.

🍷 Related Work: Linearity + Latent Dimensions ---

A key observation across deep learning: **concepts tend to be linear directions** in representation spaces.

- **word2vec** (Mikolov et al., 2013): $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen} \rightarrow$ “gender” is a linear direction
- **Probing classifiers** (Alain & Bengio, 2016): linear classifiers on hidden layers extract semantic information
- **Network Dissection** (Bau et al., 2017): individual units align with concepts — the axes themselves are meaningful

TCAV builds on this

If concepts are **linear directions** in activation space, we can find them with a simple linear classifier and measure their influence via **directional derivatives**.

Approach: Defining the CAV ---

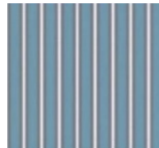
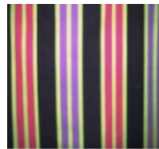
The user wants to test concept C (e.g., “stripes”).

How do we represent it inside the network?

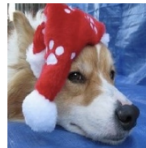
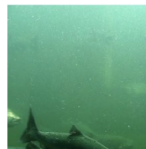
- 1 Collect a set of examples P_C of the that concept and a negative set N of examples that don't
- 2 Feed both through the network, extract activations at layer l
- 3 Train a **linear classifier** to separate them
- 4 The **CAV** $\mathbf{v}_C^l$ is the vector **orthogonal** to the decision boundary between activations — it points in the “direction of the concept”

Key insight

A CAV is a **direction** in activation space that represents a human-defined concept. No retraining of the original model needed.

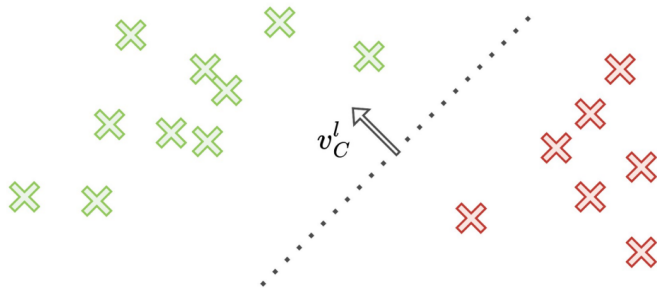


Stripes P_C



Not stripes N

🍷 Approach: Visualizing the CAV ---



The CAV $\mathbf{v}_C^l$ is the normal to the linear decision boundary:

- One side $\rightarrow$ “has concept C ”
- Other side $\rightarrow$ “does not have concept C ”

Moving along the CAV direction = “adding more of concept C ” to the representation.

$$\text{green } \times \in \{f_l(x) : x \in P_C\}$$

$$\text{red } \times \in \{f_l(x) : x \in N\}$$

Approach: Concept Sensitivity ---

Saliency maps gauge sensitivity with respect to per-pixel perturbations. With CAV, we gauge sensitivity **towards a concept** at an entire layer:

$$S_{C,k,l}(\mathbf{x}) = \nabla h_{l,k}(f_l(\mathbf{x})) \cdot \mathbf{v}_C^l$$

In plain English

“If we nudge the activations in the direction of concept C , how much does the prediction for class k change?”

- $f_l(\mathbf{x})$: activations at layer l for input $\mathbf{x}$
- $\nabla h_{l,k}(f_l(\mathbf{x}))$: gradient of class k output w.r.t. those activations
- $\mathbf{v}_C^l$: the CAV (concept direction)
- The **dot product** measures alignment between the gradient and the concept direction

Approach: The TCAV Score ---

The sensitivity $S_{C,k,l}(\mathbf{x})$ is **per-image**. To get a **global** measure for the entire class:

$$\text{TCAV}_{Q_{C,k,l}} = \frac{|\{\mathbf{x} \in X_k : S_{C,k,l}(\mathbf{x}) > 0\}|}{|X_k|}$$

In plain English

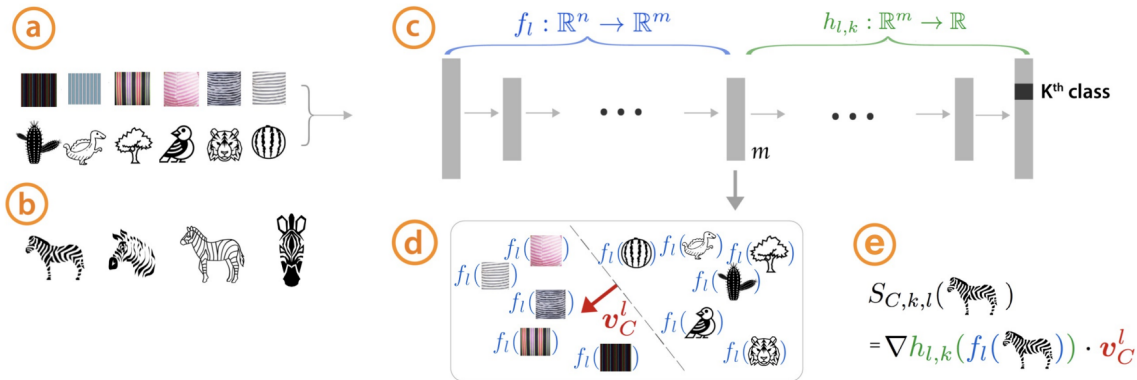
“Of all images of class k , what fraction has activations that are **positively influenced** by concept C ?”

- $\text{TCAV} = 1.0 \rightarrow$ every image of class k is sensitive to C
- $\text{TCAV} = 0.5 \rightarrow$ no more influence than random (meaningless)
- $\text{TCAV} = 0.0 \rightarrow$ concept C pushes **away** from class k

Statistical safeguard

A t -test across multiple random negative sets filters spurious CAVs. Only statistically significant results are reported.

🍷 Approach Summary ---



The full pipeline in one sentence

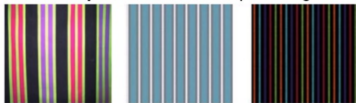
User provides concept examples $\rightarrow$ linear classifier on activations $\rightarrow$ CAV direction $\rightarrow$ directional derivative $\rightarrow$ TCAV score $\rightarrow$ t -test. **The original model is never modified.**

🍷 Results: Sorting Images with CAVs ---

Images sorted by their projection onto a CAV direction, a qualitative validation that the CAV captures the intended concept.

- Class “stripes” sorted by concept “CEO”: confirms CAV reflects the concept correctly
- Class “necktie” sorted by concept “model woman”: reveals a gender bias in the learned representation

CEO concept: most similar striped images



Model Women concept: most similar necktie images



CEO concept: least similar striped images



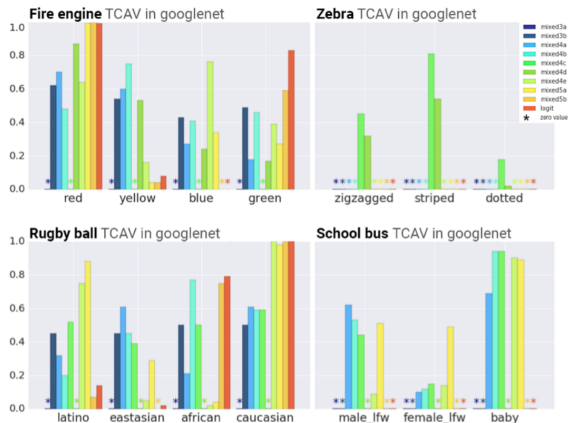
Model Women concept: least similar necktie images



Sorting by unexpected concepts can reveal biases invisible to standard evaluation metrics.



Results: Gaining Insights with TCAV ---



Each bar = TCAV score at a different layer.

Only statistically significant CAVs shown.

- “Red” → “fire engine”: high TCAV
- “Striped” → “zebra”: high TCAV
- “Dotted” → “zebra”: filtered out by t -test
- “Caucasian” → “rugby ball”: high TCAV
- ► TCAV enables **ranking concepts by importance** for a class — not just “is it relevant?” but “how much more relevant than others?”

Layers closer to the output have greater influence on the prediction.

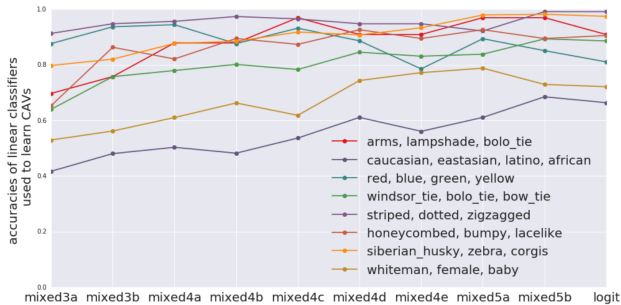
Key result

TCAV confirms intuitive associations **and** reveals hidden biases — with statistical rigor.

🍷 Results: TCAV for Where Concepts Are Learned ---

CAV accuracy at different layers reveals *where* each concept is learned:

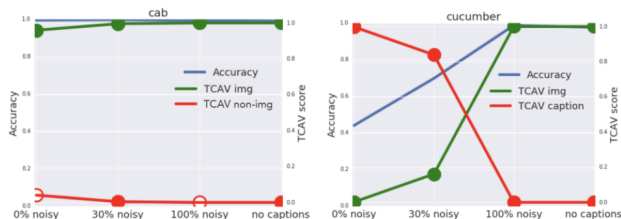
- Simple concepts (colors, patterns) reach high accuracy at low layers.
- Complex concepts (age, sex, objects) don't reach high accuracy until higher layers.



Convergence with Network Dissection

Both methods confirm the same hierarchy using completely different approaches: early layers = low-level features, late layers = high-level concepts.

🍌 Results: Controlled Experiment with Ground Truth ---



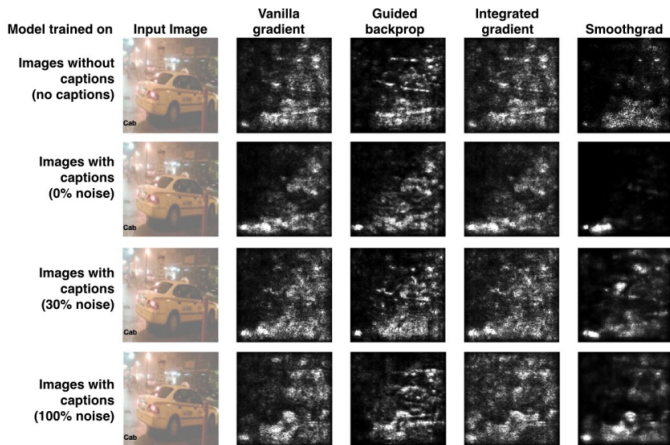
A model trained on images with **embedded captions** to create known ground truth:

- **“Cab”**: image concept dominates → TCAV correctly shows high image sensitivity, low caption sensitivity
- **“Cucumber”**: caption concept dominates when present → TCAV correctly reflects this

Validation

TCAV scores faithfully reflect the **true** importance of each concept — confirmed against known ground truth.

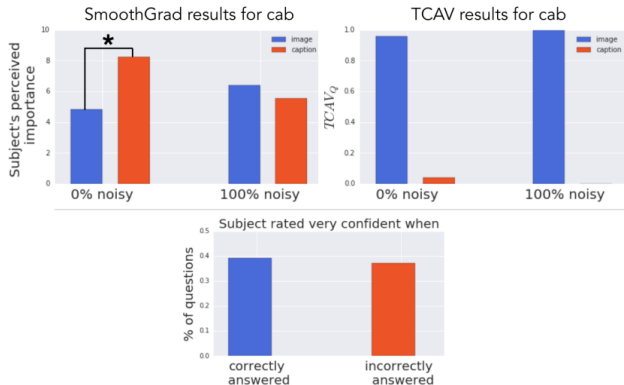
🍷 Results: Evaluation of Saliency Maps ---



We know that for “cab” the image concept is most important. But saliency maps highlight the caption text (high contrast) instead — across all four methods.



Results: Saliency Maps Are Misleading ---



Using the same controlled experiment:

- **Saliency maps** led human subjects to **incorrect conclusions** about which concept was more important
- Subjects thought the caption was more important than the image for “cab” — **wrong**
- **TCAV scores** correctly identified the dominant concept in every case

Saliency maps are not just imprecise — they are actively **misleading**. They highlight visually salient pixels (high contrast text), not causally important ones.



Results: TCAV for Medical Diagnosis ---

TCAV applied to a diabetic retinopathy diagnosis model:

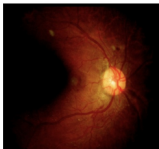
- **Green** = medically relevant concepts
- **Red** = irrelevant concepts
- **Level 4**: model correctly uses relevant concepts
- **Level 1**: model gives too much importance to HMA (relevant for Level 2, not Level 1)

Debugging with TCAV

A doctor (no ML expertise) can verify whether the model uses the **right medical concepts** — not just whether it gets the right answer.

- TCAV score shows model successfully distinguishes relevant and irrelevant concepts for level 4 DR diagnosis
- TCAV score reveals model gives too much

DR level 4 Retina



TCAV for DR level 4



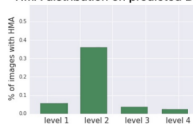
DR level 1 Retina



TCAV for DR level 1



HMA distribution on predicted DR



Conclusions ---

Roadmap: Gradient $\rightarrow$ Attention $\rightarrow$ Concept-based (**TCAV!**)

Limitations:

- Evaluated only on computer vision tasks
- Assumes concepts are **linear** directions in activation space
- Statistical significance testing — is a t -test rigorous enough?
- Depends on quality of user-provided concept examples

🍷 Discussion Questions (Paper 2) ---

- How does this method compare to e.g., saliency maps?
- How would you adapt TCAV for other types of data (e.g., audio/video) and what would that look like?
- Thoughts on TCAV for adversarial example identification (Appendix A)?
- Do you think there could be adversarial images that allow meaningless CAVs to pass the statistical significance test?

Thank You!

References --- |